

**MULTI-AGENT URBAN TRAFFIC SIGNAL
OPTIMIZATION IN SIMULATED ENVIRONMENTS**

Project Id: 25-26J-523

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in Software
Engineering

Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology

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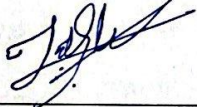
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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Signature of the supervisor:

Date: 20-08-25

ABSTRACT

This research component focuses on developing an innovative real-time urban traffic simulation and interactive control panel as part of a comprehensive multi-agent traffic signal optimization system. The component addresses critical limitations in existing traffic simulation platforms by introducing dynamic parameter adjustment capabilities, unified web-based control interfaces to start, pause or stop the simulation, scenario switching and rollback to saved states. The purpose is to enhance multi-agent traffic signal optimization by providing an intuitive interface for experimenting with different traffic scenarios without manual scripting. It draws on research into integrated simulation platforms, web-based visualization, and interactive simulation data management. The research integrates modern web technologies with established traffic simulation platforms such as Simulation of Urban Mobility (SUMO), CityFlow, and Verkehr In Städten – SIMulationsmodell (VISSIM) through standardized RESTful APIs, enabling seamless real-time interaction and control. This work contributes to the broader objective of developing AI-driven traffic signal optimization systems that can adapt to real-time traffic conditions while providing intuitive interfaces for researchers, traffic engineers, and system operators.

Keywords— Real-time traffic simulation, multi-agent systems, traffic signal optimization, web-based control interface, scenario management, integrated simulation platforms, RESTful APIs, interactive data visualization.

TABLE OF CONTENTS

DECLARATION	i
ABSTRACT	ii
LIST OF FIGURES	iv
LIST OF ABBRIVIATIONS	v
1. INTRODUCTION	1
1.1 Background and Literature Review	2
1.2 Research Gap	3
1.3 Research Problem.....	6
2. OBJECTIVES	7
2.1 Main Objectives	7
2.2 Specific Objectives.....	7
3. METHODOLOGY	10
4. PROJECT REQUIREMENTS	15
5. REFERENCES.....	19

LIST OF FIGURES

Figure 3.1: Overall System Diagram	11
Figure 3.2: System Diagram for Traffic Simulation and Control Panel	12

LIST OF ABBRIVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
COM	Component Object Model
CSV	Comma-Separated Values
ELK	Elasticsearch, Logstash, and Kibana
GNN	Graph Neural Network
GUI	Graphical User Interface
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
IEEE	Institute of Electrical and Electronics Engineers
JSON	JavaScript Object Notation
MPC	Model Predictive Control
REST	Representational State Transfer
RESTful	Representational State Transfer-compliant
RL	Reinforcement Learning
SUMO	Simulation of Urban MObility
TraCI	Traffic Control Interface
UI	User Interface
VISSIM	Verkehr In Städten - SIMulationsmodell (Traffic in Cities - Simulation Model)
XML	eXtensible Markup Language

1. INTRODUCTION

Urban traffic congestion remains a persistent challenge for modern cities, significantly affecting commute times, fuel consumption, and air quality [3]. Traditional signal control systems typically use fixed-time or simple actuated logic that lacks responsiveness to real time traffic fluctuations [3], [2]. Such static controls cannot adapt to peak demand or handle non-recurring events like accidents and roadworks [5], leading to localized congestion and broader network inefficiencies. Moreover, critical functions like emergency vehicle preemption are often implemented via simplistic overrides that neglect network-wide effects [7]. These limitations are especially problematic in rapidly urbanizing regions with limited infrastructure, where inefficiencies are exacerbated [8].

To address these issues, researchers advocate data-driven adaptive control and simulation-based analysis. Modern traffic simulation platforms like SUMO [4], CityFlow [5], VISSIM [6], and Aimsun enable virtual testing of adaptive strategies in large networks [9]. SUMO, for instance, is an open-source microscopic simulator designed for large, multimodal networks [10]. It supports real-time interaction via the Traffic Control Interface (TraCI) [10]. CityFlow is a high-performance simulator optimized for large-scale and multi-agent scenarios, offering multithreading and interactive visualization [11]. Commercial tools like VISSIM and Aimsun provide rich Graphical User Interfaces (GUIs) and Application Programming Interfaces (APIs) for control, but these often require proprietary software and still lack unified external control [12]. Although these platforms offer detailed modeling and custom APIs [9], [10], they exhibit important shortcomings. Real-time interactions are typically script-driven and unintuitive [9], no single interface unifies simulation control [9], and switching scenarios requires full restarts [9]. Existing dashboard projects improve visualization of simulation outputs [3],[2], but they often decouple the simulator and do not support real time interactive control.

In summary, the state of the art offers powerful simulation engines and separate visualization tools, but lacks an integrated, user-friendly environment for interactive, real time simulation control and analysis. This research aims to bridge that gap. Building on the project’s multi-agent optimization framework, we will implement a dedicated Simulation Control Panel. Which is a web-based interface that centrally controls and monitors the traffic simulator [9]. By combining a live simulator backend with a modern RESTful User Interface (UI), the system will enable users to start, pause, and modify simulations real time, switch scenarios without restarting, and roll back to prior states [9]. This approach aims to overcome current limitations of simulation environments and provide enhanced accessibility and flexibility needed for real-time traffic signal optimization.

1.1 Background and Literature Review

Modern traffic simulators have evolved to model individual driver behavior and support large networks. SUMO is a portable, open-source simulator that allows online control of all simulation objects via its TraCI interface [10]. VISSIM offers a high-fidelity 2D or 3D GUI and Component Object Model (COM) APIs, making it user-friendly for network editing [12], but it is commercial and less extensible. Newer tools like CityFlow focus on multi-agent reinforcement learning. CityFlow can simulate city-sized networks up to 20 times faster than SUMO and includes an interactive renderer for monitoring [11]. It allows flexible scenario definition for Reinforcement Learning (RL) experiments, though its primary focus is algorithmic control rather than a general user interface.

Several recent studies integrate simulation with visualization. A web-based application for managing and visualizing large-scale traffic simulation outputs [3], and an open-source real-time visualization platform linking VISSIM simulation with an Elasticsearch, Logstash, and Kibana (ELK) stack [2] has been developed.

However current traffic simulation platforms exhibit significant limitations in real-time interaction capabilities. While simulators like SUMO support real-time signal control via

TraCI, they require manual coding or scripting to adjust parameters during runtime [3]. The absence of intuitive, visual interfaces makes real time interaction less accessible to users unfamiliar with simulation backend systems.

Existing platforms lack unified control interfaces for comprehensive simulation management. Most simulators rely on external scripts or internal GUIs that provide minimal customization and advanced control capabilities. This limitation hinders the development of integrated systems that can seamlessly coordinate multiple simulation components.

Static scenario management represents another significant constraint in current implementations. Existing simulation platforms typically require full reset or restart to switch between different traffic scenarios, with minimal support for dynamic scenario switching or state rollback during testing or demonstration phases [1]. In summary, existing work has advanced traffic visualization and dashboarding but typically separates simulation execution from user interaction.

1.2 Research Gap

Despite significant advances in traffic simulation and control technologies, several critical gaps persist in current implementations. A comprehensive analysis of existing systems reveals multiple limitations that this research addresses.

The Traditional SUMO GUI provides basic play/pause functionality for simulation control, allowing users to start and stop simulations through its desktop interface [10]. However, this basic control lacks comprehensive real-time parameter adjustment capabilities during simulation execution, limiting users to predefined simulation configurations without dynamic experimentation possibilities. The Desktop Integration Platform offers limited basic interaction capabilities [1], but these interactions are constrained by the desktop environment and do not provide the flexibility needed for comprehensive real-time control scenarios.

The ELK Stack Visualization Platform provides web-based access for displaying simulation outputs and real-time data visualization [2]. The Web Traffic Data

Management system offers web-based interfaces for post-simulation analysis and data management [3]. While these platforms successfully demonstrate web-based accessibility for data visualization, they do not extend this accessibility to simulation control functions, creating a separation between visualization capabilities and interactive control requirements.

The VISSIM COM Interface supports parameter adjustment capabilities through its Component Object Model programming interface [6]. However, this approach requires extensive scripting knowledge and technical expertise, making it inaccessible to non-technical users. The interface lacks user-friendly methods for real-time parameter modification, requiring users to write custom scripts for each adjustment scenario.

Current analysis reveals that none of the existing systems provide adequate scenario switching or state rollback capabilities. Traditional simulation platforms, including SUMO GUI [10] and VISSIM COM Interface [6], require complete simulation restarts to switch between different traffic scenarios, preventing seamless comparative analysis and iterative testing workflows.

Table 1.1 : Existing Systems Comparisons

Feature	[2]	[3]	[1]	[10]	[6]	Proposed System
Real-time Control	No.	No.	Limited Basic interaction	Yes - Basic Play/Pause	No.	Yes
Web-based Interface	Yes	Yes	No	No	No	Yes
Dynamic Parameter Adjustment	No	No	No	No	Yes Via Scripting	Yes
Scenario Switching	No	No	No	Requires Restart	Requires Restart	Yes
State Rollback	No	No	No	No	No	Yes
Unified Control Interface	No	No	No	No	No	Yes
RESTful API Integration	No API	No API	No API	TraCI Only	COM Only	Yes

1.3 Research Problem

The core research problem addresses the development of a unified, web-based control panel that enables real-time interaction with traffic simulation platforms while supporting dynamic scenario management and rollback capabilities. This component aims to bridge the gap between theoretical simulation capabilities and practical real-time control requirements.

Summary of Key Problems:

- Lack of real-time interactive control without manual scripting requirements.
- Absence of unified web-based interfaces for accessible simulation management.
- Static scenario management requiring full simulation restarts for scenario changes.
- Limited state rollback capabilities preventing iterative experimentation.
- Poor integration between visualization and control in existing platforms.
- Technical barriers preventing non-expert users from effectively utilizing simulation tools.

2. OBJECTIVES

2.1 Main Objectives

Develop and implement a comprehensive real-time urban traffic simulation control panel that integrates with traffic simulation platforms through standardized APIs, enabling dynamic parameter adjustment, unified web-based control interfaces, and advanced scenario management capabilities with rollback functionality to support AI-driven traffic signal optimization research and practical deployment scenarios. The system will be designed with a flexible architecture that allows for potential integration with different simulation platforms while maintaining consistent functionality and user experience.

2.2 Specific Objectives

Below mentioned specific objectives must be fulfilled to achieve the overall objective described above.

Simulation Platform Integration and Setup

This objective focuses on establishing a robust foundation for the simulation environment by conducting a comprehensive comparative analysis of existing traffic simulators including SUMO, CityFlow, and VISSIM using predefined evaluation criteria such as API capabilities, performance metrics, and integration flexibility. The research will design and implement a flexible integration architecture for the selected simulation platform, ensuring adaptability and scalability. Additionally, the selected simulation platform will be configured and optimized to handle realistic urban traffic scenarios, providing an authentic testing environment for the control panel development.

Real-time Control System Development

This objective aims to create the core infrastructure that enables seamless interaction between users and the simulation environment. The development includes creating comprehensive RESTful APIs that facilitate simulation control and enable real-time data

exchange between the frontend interface and the simulation backend. Essential simulation control functionalities such as start, pause, stop, and reset operations will be implemented to provide users with complete simulation lifecycle management. Furthermore, dynamic parameter adjustment capabilities will be developed to allow real-time modification of critical simulation parameters including traffic flow rates, signal timings, and individual vehicle behaviors during simulation execution.

Web-based Control Panel Creation

This objective focuses on developing an intuitive and accessible user interface that serves as the primary interaction point between users and the simulation system. The development involves designing and implementing a modern, responsive web-based control interface using contemporary frameworks and technologies to ensure cross-platform compatibility and optimal user experience. Real-time monitoring and visualization capabilities will be integrated to display live traffic simulation data through interactive charts, graphs, and visual representations. The interface will be designed with accessibility principles in mind, ensuring that both technical and non-technical users can effectively interact with the system without requiring extensive technical knowledge.

Advanced Scenario Management System

This objective addresses the need for flexible and dynamic simulation scenario handling through the development of a comprehensive scenario management system that supports real-time switching between different predefined traffic conditions without requiring simulation restarts. The system will incorporate sophisticated state capture and rollback capabilities, enabling users to save simulation states at specific points and return to them for comparative analysis and testing purposes. Additionally, standardized scenario definition formats using structured data formats such as JavaScript Object Notation (JSON) and eXtensible Markup Language (XML) will be established to ensure consistency, portability, and ease of scenario sharing across different research contexts.

System Integration and Testing

This objective ensures the reliability, performance, and scalability of the complete system through comprehensive integration and validation processes. All developed components will be integrated into a unified, scalable system architecture that maintains modularity while ensuring seamless communication between different system layers. A comprehensive testing framework will be implemented covering both unit-level testing of individual components and integration testing of the complete system to identify and resolve potential issues. Performance optimization strategies will be developed and implemented to ensure the system operates efficiently under real-time constraints and can handle varying loads and simulation complexities.

Documentation and Evaluation

This objective focuses on ensuring long-term usability, maintainability, and validation of the developed system through comprehensive documentation and evaluation processes. Detailed user documentation and system administration guides will be created to support both end-users and system administrators in effectively utilizing and maintaining the system. Performance evaluation and usability testing will be conducted with representative target users to assess the system's effectiveness, identify areas for improvement, and validate that the system meets user requirements and expectations. Additionally, comprehensive testing scenarios will be designed and executed to validate the system's capabilities under various conditions and use cases, ensuring robustness and reliability in practical applications.

3. METHODOLOGY

The research methodology follows a systematic approach integrating software engineering principles with real-time traffic simulation requirements. The methodology is structured around developing a comprehensive web-based control panel that interfaces with microscopic traffic simulation platforms through standardized APIs, enabling real-time interaction, dynamic parameter adjustment, and advanced scenario management capabilities.

Overall System Architecture

The proposed multi-agent traffic signal optimization system adopts a modular, distributed architecture comprising four interconnected components as illustrated in Figure 3.1. The system integrates simulation environments with AI-driven optimization modules and real-time control interfaces to create a comprehensive traffic management solution.

The overall system operates through coordinated interaction between specialized modules. The Simulation and Control Panel serves as the central hub, providing the foundation environment where all other components operate. The Graph Neural Network (GNN) based optimizer and Model Predictive Control (MPC) controller function as alternative optimization strategies, while the emergency vehicle preemption system operates as an overlay service that can override normal traffic control when emergency situations arise.

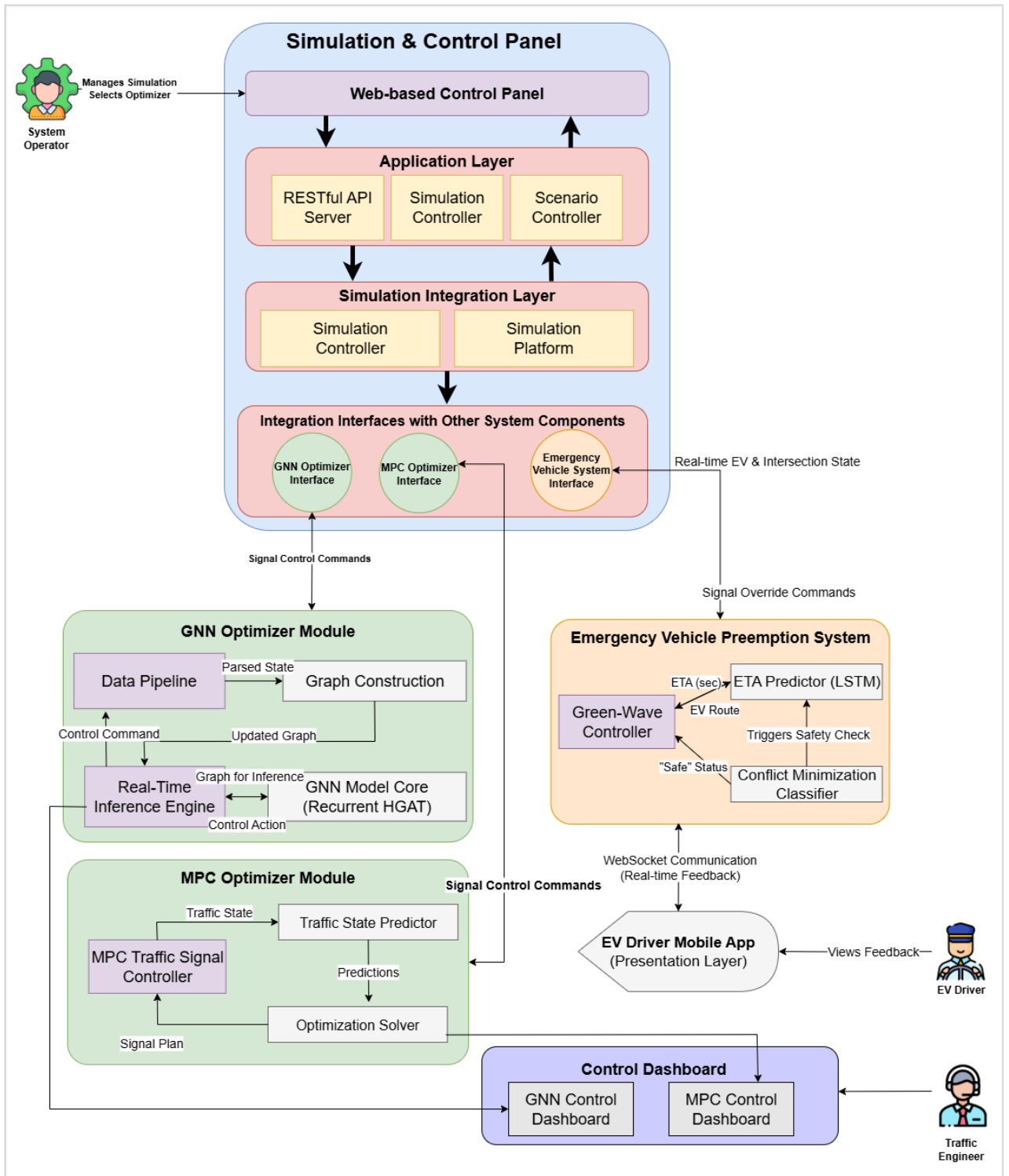


Figure 3.1: Overall System Diagram

Component Architecture - Real-Time Traffic Simulation and Control Panel

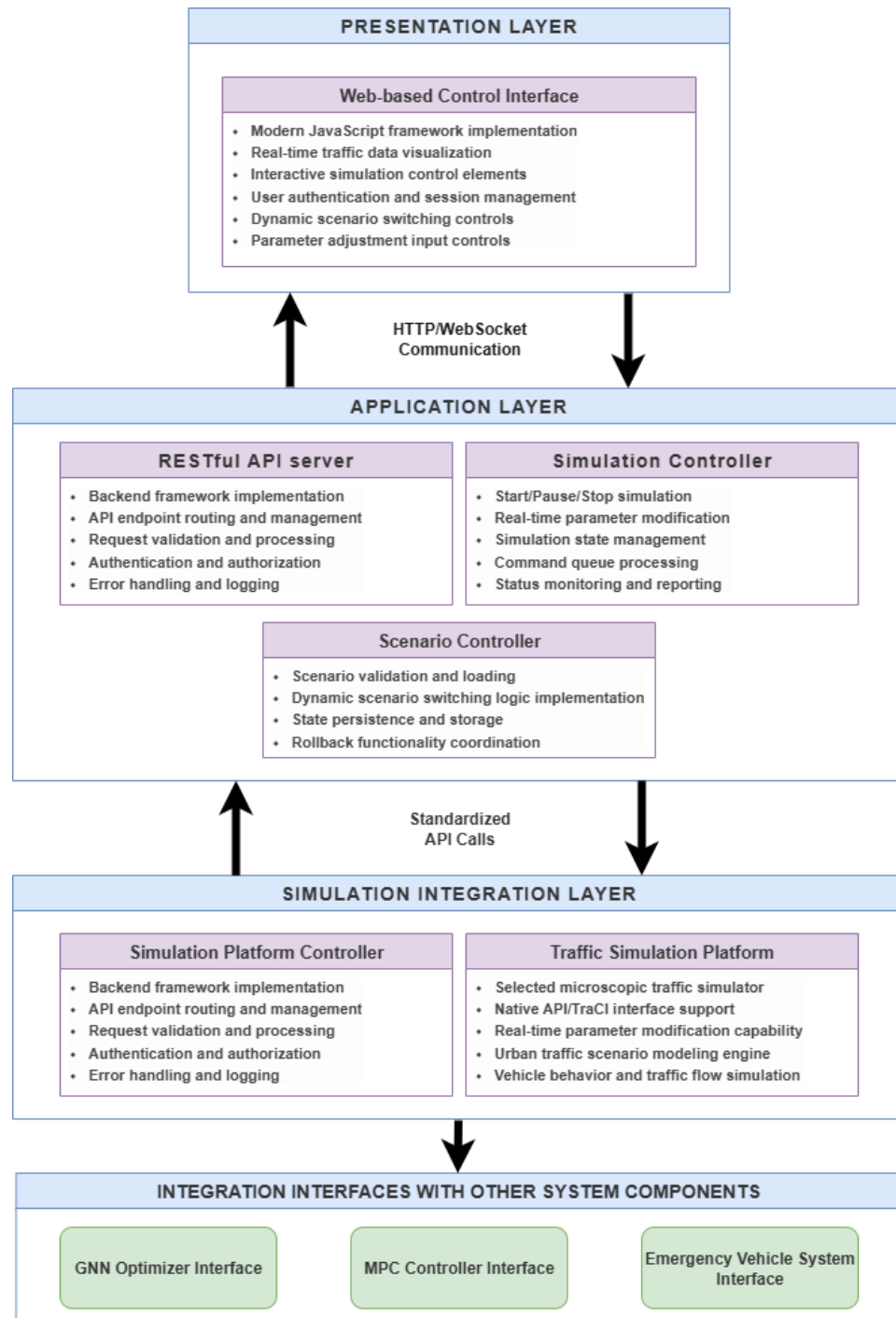


Figure 3.2: System Diagram for Traffic Simulation and Control Panel

For the Real-Time Urban Traffic Simulation and Interactive Control Panel component, the system adopts three-tier architecture comprising presentation, application, and simulation layers. This architecture ensures clear separation of concerns while enabling seamless real-time interaction between users and the simulation environment.

Presentation Layer: The presentation layer delivers a responsive web-based interface built with modern JavaScript frameworks. This layer provides real-time data visualization capabilities, interactive control elements for simulation management, and intuitive scenario switching interfaces. The interface incorporates WebSocket connections for real-time bidirectional communication with the backend services, ensuring low-latency updates and responsive user interactions.

Application Layer: The application layer implements RESTful API services using robust backend frameworks. This layer handles business logic including simulation control orchestration, scenario management and validation, state persistence and rollback mechanisms, and real-time data processing and transformation. The API services provide standardized endpoints for simulation control operations, parameter adjustment capabilities, and comprehensive data retrieval functions. This layer acts as an abstraction between the frontend interface and the simulation platform, ensuring maintainable and scalable system design.

Simulation Layer: The simulation integration layer interfaces with the selected microscopic traffic simulation platform through its native API. The system will implement a standardized adapter pattern to ensure clean abstraction and facilitate potential future platform changes if required. This layer manages direct communication with the simulation engine, real-time data collection and processing, and dynamic parameter adjustment during simulation runtime.

To address the identified research problems, this research aims to implement several key features as part of the comprehensive methodology:

Unified Web-based Control Interface

The control panel will provide a modern web interface that enables users to start, pause, and stop simulations seamlessly. The interface will support dynamic parameter adjustment

during simulation runtime, load and switch between predefined traffic scenarios and save and restore simulation states for comparative analysis and iterative testing. Real-time data visualization will be integrated through interactive charts and graphs, providing immediate feedback on simulation performance. The interface will be designed for accessibility, ensuring both technical and non-technical users can effectively interact with the system without requiring programming expertise.

Standardized API Integration Architecture

The control panel will communicate with the simulation backend through a well-defined RESTful API architecture. This abstraction layer will enable clear separation between the user interface and simulation logic, improving system scalability and maintainability while supporting future platform integrations.

Advanced Simulation Management Features

The system will implement scenario and state management capabilities to support flexible experimentation and iterative testing. Real-time scenario switching will enable transitions between different predefined traffic conditions without simulation restarts. State capture and rollback functionality will allow users to save and restore simulation states for comparative analysis. Standardized scenario definition formats using JSON and XML will ensure consistency and portability. The system will include comprehensive logging, data export capabilities in multiple formats, and real-time performance monitoring with alerting.

4. PROJECT REQUIREMENTS

Functional Requirements

- Real-time Simulation Control - The control panel will provide a modern web interface that allows users to start, pause, stop, and reset simulations. It will enable dynamic changes to input parameters such as traffic flow rates, signal timings, and vehicle behaviors. Users will also be able to load and switch between predefined traffic scenarios, as well as save and restore simulation states for rollback and comparative analysis.
- Simulator Integration - The system will integrate with a traffic simulation platform using standardized API interfaces. The architecture will remain adaptable, allowing additional platforms to be incorporated in the future if required.
- Scenario Management - The system should support the users to switch between predefined traffic scenarios without manual scripting or restarting the simulation, and the system should provide state capture and rollback capabilities to enable comparative analysis of different configurations.
- Data Export and Reporting - Simulation results should be exportable in widely used formats such as Comma-Separated Values (CSV), JSON, and XML for external processing and analysis. The system will also provide real-time performance monitoring with alerting capabilities to notify users of critical events or anomalies. In addition, a comprehensive logging mechanism will record user actions and system events to ensure transparency and traceability.

Non-Functional Requirements

- Performance Requirements - The system is designed to support real-time interaction, ensuring smooth and responsive operation during use. It must be capable of scaling to accommodate complex and large-scale urban networks, allowing the simulation to remain efficient even as the environment grows in size and complexity. In addition, the system should make use of memory resources effectively so that extended simulation runs can be conducted without performance degradation.

- Reliability and Availability - The system should remain available during operational periods, ensuring consistent access for users. It must incorporate fault-tolerance mechanisms to continue functioning effectively even in the event of component failures within the simulation platform. Furthermore, the system must preserve data integrity when users switch between different scenarios or perform rollback operations, maintaining the accuracy and consistency of simulation outcomes.
- Security Requirements - The system must include secure authentication and authorization mechanisms to ensure that only authorized users are able to access and control its features. Measures should be in place to protect against any form of unauthorized access to critical simulation functions. All data transmitted within the system should be safeguarded through secure communication protocols, protecting sensitive information from interception or tampering.
- Usability Requirements - The system should provide an intuitive user interface that enables users to perform basic operations with minimal training. Comprehensive user guides and help documentation must be available to support both novice and advanced users. In addition, the system should comply with accessibility standards so that users with disabilities are able to engage with the platform effectively and without barriers.
- Maintainability Requirements - The system should follow modular architecture that allows components to be updated or replaced independently, reducing the complexity of maintenance tasks. A strong testing framework must be in place to ensure that both unit and integration testing provide thorough coverage of the system. Clear and detailed documentation should be available for both system administrators and developers, supporting efficient maintenance, troubleshooting, and future enhancements.

User requirements

- The system should provide an intuitive web-based interface that allows users to control simulations in real-time without requiring extensive technical knowledge or manual scripting. Users should be able to start, pause, stop, and modify simulation parameters seamlessly through visual controls.
- Researchers and traffic engineers should have access to comprehensive scenario management capabilities, enabling them to switch between different predefined traffic scenarios during simulation runtime. The system should support rollback functionality, allowing users to return to previous simulation states for comparative analysis.
- System developers and AI researchers should receive standardized API access for integrating their optimization algorithms and control systems. The platform should provide real-time data streaming capabilities that support automated testing and algorithm validation workflows.
- Urban planners and educational users should have access to simplified visualization tools and pre-configured scenarios that require minimal setup. The system should generate exportable reports and data in standard formats (CSV, JSON, XML) for further analysis and presentation purposes.

The design process will focus on creating a responsive, accessible interface that supports cross-platform compatibility and real-time data visualization. Special attention will be given to the scenario switching system, ensuring that transitions between different traffic conditions are smooth and do not interrupt the simulation flow. Prototypes will be developed using modern web frameworks, allowing for rapid iteration and user testing. These prototypes will be evaluated with target user groups including traffic researchers

and urban planners for feedback, ensuring that the interface meets diverse user needs and supports both technical and non-technical users effectively.

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